Matrix Arithmetic

We can perform element by element mathematical operations on a matrix with a scalar (single number) just like we could with vectors. Let's see some quick examples:

In [3]:

```
mat <- matrix(1:50,byrow=TRUE,nrow=5)
mat
```

Out[3]:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

In [2]:

```
# Multiplication
2*mat
```

Out[2]:

2	4	6	8	10	12	14	16	18	20
22	24	26	28	30	32	34	36	38	40
42	44	46	48	50	52	54	56	58	60
62	64	66	68	70	72	74	76	78	80
82	84	86	88	90	92	94	96	98	100

In [4]:

```
# Division (reciprocal)
1/mat
```

Out[4]:

1.0000000	0.5000000	0.3333333	0.2500000	0.2000000	0.1666667	0.1428571	0.1250000	0.1111111	0.1000000
0.09090909	0.08333333	0.07692308	0.07142857	0.06666667	0.06250000	0.05882353	0.05555556	0.05263158	0.0500000
0.04761905	0.04545455	0.04347826	0.04166667	0.04000000	0.03846154	0.03703704	0.03571429	0.03448276	0.0333333
0.03225806	0.03125000	0.03030303	0.02941176	0.02857143	0.02777778	0.02702703	0.02631579	0.02564103	0.0250000
0.02439024	0.02380952	0.02325581	0.02272727	0.02222222	0.02173913	0.02127660	0.02083333	0.02040816	0.0200000

In [5]:

```
# Division
mat/2
```

Out[5]:

0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0

5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0
10.5	11.0	11.5	12.0	12.5	13.0	13.5	14.0	14.5	15.0
15.5	16.0	16.5	17.0	17.5	18.0	18.5	19.0	19.5	20.0
20.5	21.0	21.5	22.0	22.5	23.0	23.5	24.0	24.5	25.0

In [8]:

```
# Power
mat ^ 2
```

Out[8]:

1	4	9	16	25	36	49	64	81	100
121	144	169	196	225	256	289	324	361	400
441	484	529	576	625	676	729	784	841	900
961	1024	1089	1156	1225	1296	1369	1444	1521	1600
1681	1764	1849	1936	2025	2116	2209	2304	2401	2500

Comparison operators with matrices

We can similarly perform comparison operations across an entire matrix to return a matrix of logicals:

In [9]:

```
mat > 17
```

Out[9]:

FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

Matrix Arithmetic with multiple matrices

We can use multiple matrices with arithmetic as well, for example:

In [13]:

```
mat + mat
```

Out[13]:

2	4	6	8	10	12	14	16	18	20
22	24	26	28	30	32	34	36	38	40
42	44	46	48	50	52	54	56	58	60
62	64	66	68	70	72	74	76	78	80
82	84	86	88	90	92	94	96	98	100

In [14]:

```
mat / mat
```

```
mat / mat
```

Out[14]:

1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1

In [16]:

```
# Warning, big numbers!
mat ^ mat
```

Out[16]:

1	4	27	256	3125	46656	823543	16777216	387
2.853117e+11	8.916100e+12	3.028751e+14	1.111201e+16	4.378939e+17	1.844674e+19	8.272403e+20	3.934641e+22	1.9
5.842587e+27	3.414279e+29	2.088047e+31	1.333736e+33	8.881784e+34	6.156120e+36	4.434265e+38	3.314552e+40	2.5
1.706917e+46	1.461502e+48	1.291100e+50	1.175664e+52	1.102507e+54	1.063874e+56	1.055513e+58	1.075912e+60	1.1
1.330878e+66	1.501309e+68	1.734377e+70	2.050774e+72	2.480636e+74	3.068035e+76	3.877924e+78	5.007021e+80	6.6

In [17]:

```
mat*mat
```

Out[17]:

1	4	9	16	25	36	49	64	81	100
121	144	169	196	225	256	289	324	361	400
441	484	529	576	625	676	729	784	841	900
961	1024	1089	1156	1225	1296	1369	1444	1521	1600
1681	1764	1849	1936	2025	2116	2209	2304	2401	2500

Matrix multiplication

A quick side note on matrix multiplications. You can perform arithmetic multiplication on an element by element basis using the * sign in R. You should note this is not the same as Matrix Multiplication. If you are familiar with the mathematics behind this topic and want to use R to perform true matrix multiplication, you can use the following:

In [18]:

```
mat2 <- matrix(1:9, nrow=3)
```

In [19]:

```
mat2
```

Out[19]:

1	4	7
2	5	8
3	6	9

In [20]:

```
# True Matrix Multiplication
mat2 %*% mat2
```

Out[20]:

30	66	102
36	81	126
42	96	150

Okay, that's it for matrices! Later on we will learn how to use **factor()** to quickly create categoical matrices from character elements.